

TRINIDAD STOKES, CO, USA

WMO: 724645

Lat: 37.262N

Lon: 104.338W

Elev: 1750

StdP: 81.99

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23070

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-15.7	-12.6	-21.7	0.7	-3.5	-19.0	0.9	-1.0	14.3	8.2	12.0	8.3	3.0	270	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.5	34.5	15.2	33.1	15.0	31.9	15.0	18.0	27.2	17.3	27.0	16.8	26.5	4.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	13.8	19.8	14.6	12.9	19.7	13.8	12.2	19.6	57.6	26.8	55.4	26.7	53.5	26.6	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.1	9.3	-21.8	36.9	3.7	1.2	-24.5	37.7	-26.7	38.4	-28.8	39.0	-31.5	39.9		
(5)			-22.2	19.3	3.6	0.9	-24.8	19.9	-26.9	20.4	-28.9	20.9	-31.6	21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	0.7	2.2	6.4	10.0	15.2	21.1	23.6	22.4	18.7	11.8	5.6	0.5		
	DBStd	9.55	5.76	6.00	5.62	5.25	4.54	3.66	2.61	2.46	4.18	5.21	5.59	5.81		
	HDD10.0	1210	289	222	136	64	11	0	0	0	3	41	149	295		
	HDD18.3	2936	546	451	371	252	114	16	1	2	44	206	382	552		
	CDD10.0	1788	2	4	24	63	172	334	421	386	265	97	18	2		
	CDD18.3	472	0	0	0	1	18	99	164	129	56	4	0	0		
	CDH23.3	5922	0	1	11	59	400	1357	1843	1387	728	131	5	0		
	CDH26.7	2537	0	0	0	6	116	629	903	603	261	20	0	0		
(14) Wind	WSAvg	4.4	4.2	4.4	4.8	5.2	4.8	4.7	4.2	4.0	4.2	4.2	4.2	4.1		
(15) Precipitation	PrecAvg	344	10	13	22	27	37	37	55	52	30	23	19	14		
	PrecMax	527	33	32	66	118	124	162	166	126	75	72	60	46		
	PrecMin	222	2	1	1	0	3	4	9	5	0	0	0	0		
	PrecStd	86	8	9	16	27	27	32	37	34	17	20	14	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	21.4	24.9	27.4	32.0	36.2	36.1	34.9	33.1	28.9	24.0	18.9		
		MCWB	6.3	7.0	8.6	9.9	12.8	14.5	16.0	15.3	14.5	11.7	8.8	6.5		
	2%	DB	16.2	18.2	22.5	25.1	29.7	34.2	34.6	33.3	31.4	26.9	21.4	16.0		
		MCWB	5.3	5.8	7.7	9.1	12.2	14.2	15.7	15.4	14.1	10.9	7.9	5.1		
	5%	DB	13.3	15.9	20.1	23.0	27.6	32.5	33.3	31.9	29.8	24.5	18.8	13.3		
		MCWB	3.9	4.9	6.9	8.7	11.7	14.0	15.7	15.6	13.8	10.6	6.9	4.0		
	10%	DB	10.5	12.8	17.5	21.0	25.6	30.7	32.0	30.4	27.9	22.1	16.0	10.4		
		MCWB	2.6	3.4	5.9	8.1	11.1	13.9	15.8	15.6	13.7	9.8	5.7	2.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	7.7	9.5	11.6	15.3	17.6	18.9	18.8	17.0	14.5	9.6	7.1		
		MCDB	17.8	19.8	22.1	22.8	25.9	28.5	27.6	27.0	25.8	23.6	21.4	17.8		
	2%	WB	5.5	6.3	8.3	10.3	14.1	16.6	18.1	17.9	16.1	12.8	8.4	5.5		
		MCDB	15.2	17.5	20.6	21.5	24.6	27.5	27.4	26.8	25.4	22.2	20.0	15.0		
	5%	WB	4.1	5.1	7.3	9.4	13.1	15.8	17.5	17.3	15.3	11.5	7.3	4.3		
		MCDB	12.7	15.0	18.9	20.5	23.2	26.8	27.3	26.5	25.1	21.5	17.7	12.8		
	10%	WB	2.8	3.8	6.3	8.6	12.2	15.2	16.9	16.7	14.6	10.3	6.1	2.8		
		MCDB	10.1	12.3	16.6	19.7	22.6	26.3	26.9	25.9	24.4	20.5	15.4	10.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	15.4	16.2	15.7	15.8	16.3	15.5	15.0	15.5	16.1	15.8	14.6		
		MCDBR	18.6	19.3	19.6	19.0	19.0	18.5	17.3	17.1	17.4	18.9	19.4	18.8		
		MCWBR	10.2	10.0	9.1	8.2	7.5	6.6	4.9	5.1	5.8	7.9	9.3	10.3		
	5% WB	MCDBR	17.7	18.5	18.6	17.4	16.5	16.1	14.9	14.7	14.8	16.5	18.2	17.9		
		MCWBR	10.0	9.7	9.0	8.0	7.0	6.3	4.9	5.2	5.2	7.3	9.0	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.236	0.242	0.269	0.293	0.307	0.327	0.359	0.343	0.303	0.273	0.248	0.239		
	taud		2.629	2.632	2.547	2.481	2.503	2.451	2.435	2.466	2.546	2.608	2.605	2.606		
	Ebn at Noon		987	1016	1005	989	973	947	914	925	953	963	960	959		
	Edh at Noon		69	78	94	107	106	112	113	107	93	79	70	66		
(44) All-Sky Solar Radiation	RadAvg		2.98	3.90	5.16	6.25	7.04	7.50	7.07	6.42	5.62	4.37	3.25	2.60		
	RadStd		0.20	0.28	0.35	0.37	0.45	0.31	0.29	0.27	0.29	0.25	0.21	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.44	N/A	-1.54	+0.70	N/A	N/A	N/A	N/A	+104	+75		
(47)	Regional (16 neighbors)	+0.32	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78	+43		

Nomenclature: See separate page